ExamGuard AI

Implementation Guide & Learning Path

8 Phases | 59 Topics | From Zero to Working System

Personal Project Blueprint | March 2026

The Problem & Solution

THE PROBLEM

100+ students per hall
2-3 invigilators only
50+ halls in big exams
Humans get tired
Can't watch everyone

Cheating goes undetected
Unfair to honest students

EXAMGUARD AI

AI watches ALL cameras
ALL students, ALL the time
Never gets tired

Detects suspicious behavior
Alerts invigilator instantly
Saves evidence clips

Human makes final decision
AI assists, doesn't punish

Cameras -> AI Processes -> Detects Cheating -> Alerts Invigilator -> Human Decides

System Architecture — 5 Layers

1. VIDEO INPUT

Camera feeds
Frame extraction
Pre-processing

2. DETECTION

YOLO: find students
CNN: gaze direction
Objects: phone, chit

3. ANALYSIS

Behavior classify
Anomaly detection
Multi-student sync

4. DECISION

Confidence score
Alert threshold
Priority ranking
False alarm filter

5. OUTPUT

Dashboard
Alert notification
Evidence clips
Post-exam report

3 ML Types Working Together

SUPERVISED

WHAT is happening?

Recognize cheating
from labeled clips

YOLO: detect phone
CNN: gaze direction
CNN+LSTM: behavior

Train: 10K+ clips

UNSUPERVISED

What PATTERNS are unusual?

Learn normal behavior
Flag anything different

Autoencoder: learn normal
Isolation Forest: outliers
K-Means: group behaviors

No labels needed

REINFORCEMENT

WHEN to alert?

Balance false alarms
vs missed cheating

Correct alert: +100
False alarm: -50
Missed cheating: -200

Learns over time

Learning Roadmap — 8 Phases

Phase 1
Foundations

AI, ML, Math
Model Selection

DONE!

Phase 2
Python

Python, NumPy
Pandas, Matplotlib

3-4 wks

Phase 3
Core ML

Scikit-learn
Metrics, Imbalanced

4-6 wks

Phase 4
Deep Learning

PyTorch, CNN
Transfer Learning

6-8 wks

Phase 5
Computer Vision

OpenCV, YOLO
Face, Gaze, Pose

6-8 wks

Phase 6
Advanced ML

LSTM, Autoencoder
RL, Real-time

6-8 wks

Phase 7
System Build

Camera, Dashboard
API, Docker, GPU

8-12 wks

Phase 8
Testing

Privacy, Pilot
Optimize, Deploy

4-6 wks

ExamGuard Model Map — 7 Sub-Problems

Detect Phone
YOLO

Supervised
Pre-trained, 30+fps

Where Looking?
CNN

Supervised
Best for images

Behavior?
CNN+LSTM

Supervised
Frames + time

Unusual?
Autoencoder

Unsupervised
Learn normal

Group
K-Means

Unsupervised
Simple, proven

When Alert?
Custom RL

Reinforcement
Balance alerts

Track Person
Re-ID Net

Deep Learning
Match angles

MVP Versions — Start Small, Build Up

v0.1

Phone Detector

Single camera
Detect phones only
YOLO pre-trained

Build after Phase 5
Useful on its own!

v0.2

Gaze Tracker

Single camera
Track where looking
MediaPipe + timer

Alert: 10+ sec at neighbor

v0.3

Behavior Monitor

Single camera
Phone + Gaze + Pose
+ Anomaly detection
Confidence scoring

Build after Phase 6

v1.0

Single Hall

4-5 cameras
Full detection pipeline
Dashboard + alerts
Evidence clips

Build after Phase 7

v2.0

Multi-Hall

50+ cameras
Edge processing
Control room
Full reporting

Build after Phase 8

Multi-Camera Scaling

The Challenge

1 hall = 5 cameras = 150 fps
10 halls = 50 cameras = 1500 fps
50 halls = 250 cameras = 7500 fps

Same student from 2-3 angles

The Solution

Edge processing: 90% reduction
Smart FPS: 5-10 normally, 30 when suspicious
Person Re-ID: track across cameras
1 GPU = 20-30 camera feeds

Small

Cameras: 4-5
GPUs: 1 GPU
Students: 50-100

Medium

Cameras: 40-50
GPUs: 2-3 GPUs
Students: 500-1K

Large

Cameras: 200+
GPUs: 8-10 GPUs
Students: 5000+

Timeline — Zero to Working ExamGuard

Month 1-2	Foundations + Python	Phase 1-2
Month 3-4	Core ML + Deep Learning	Phase 3-4
Month 5-6	Computer Vision	Phase 5 (CRITICAL)
Month 6	v0.1 Phone Detector	FIRST WORKING MVP!
Month 7-8	Advanced ML	Phase 6
Month 8	v0.3 Behavior Monitor	COMBINED MVP
Month 9-11	System Building	Phase 7
Month 11	v1.0 Single Hall	REAL PRODUCT
Month 12-14	Testing + Multi-Hall	Phase 8 + v2.0